Adheesh Ramanath

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EDUCATION

Junior | **Plano West Senior High School**, Plano, TX *GPA: 4.63/5* *Rank: 18/1381*

Relevant Coursework: AP Calculus BC, AP Pre-Calculus, AP Physics 1, AP Computer Science A, AP Computer Science Principles, PLTW Digital Electronics, PLTW Engineering Science, Harvard CS50P, Harvard CS50AI

EXPERIENCE

Shining Mindz, Frisco, TX May 2025 - August 2025

Automation Engineering Intern

- Developed a custom VBA script to automate data extraction from PowerPoint into Excel, reducing workload
- Enhanced data accuracy and reporting by standardizing automated workflows across recurring project updates
- Worked with a mentor to analyze data requirements and design automation solutions aligned with operation needs
- Tested, debugged, and optimized VBA scripts to ensure reliability across large datasets and varied input formats
- Documented automation logic and workflows to support long-term maintenance and seamless team handoff

Plano West Robotics, Plano, TX August 2025 - May 2026

Robotics Engineer

- Designed and built intake and ball-sorting mechanisms for a competition robot to execute precise scoring patterns
- Integrated robot subsystems with team members to ensure reliable performance during full-system operation
- Refined designs through testing and debugging to improve alignment, timing, and mechanical consistency
- Worked with engineers to evaluate design tradeoffs and make data-driven decisions based on performance result
- Qualified for the UIL State Robotics Competition through successful competition results and design excellence

PROJECTS

GestureCanvas - AI Hand Gesture Drawing System | Python, OpenCV, MediaPipe Winter 2025

- Create a real-time hand gesture recognition system in Python using machine learning for touch-less interaction
- Implemented computer vision pipeline for hand detection, tracking, and gesture classification in real-time
- Integrated AI models to interpret dynamic gestures with high accuracy, low latency, and consistent performance
- Engineered an intuitive interaction tool with applications in digital art, accessibility, and HCI research
- Project qualified to the Dallas Regional Science and Engineering Fair through competitive judging and evaluation

Autonomous Sensor Fusion Navigation System | Arduino, C++, Embedded Systems Fall 2025

- Developed an embedded autonomous navigation system using multi-sensor input for real-time obstacle detection
- Implemented sensor fusion logic to combine distance readings and improve reliability under noisy conditions
- Designed real-time decision algorithms to trigger autonomous responses based on proximity and confidence levels
- Optimized embedded C++ code for low-latency performance on resource-constrained microcontroller hardware
- Tested system performance across repeated trials, refining thresholds to ensure stable and consistent behavior

Embedded Vision-Based Obstacle Detection System | Arduino, C++ Summer 2025

- Built Arduino-based glasses using ultrasonic sensors to detect obstacles and provide real-time audio navigation
- Documented wiring diagrams and source code to ensure reproducibility and professional project presentation
- Programmed DFPlayer Mini audio output to deliver distance-based voice prompts for safer user navigation
- Integrated sensors and microcontroller logic while optimizing programs for consistent real-time performance

Biomimicry owl and Automated Cookie Machine | VEX Robotics, C++ Winter 2024

- Built a VEX Robotics owl with head rotation and an automated cookie dispenser using C++ motor control
- Programmed smooth motion algorithms for servos and motors to enable precise, timed dispensing actions
- Integrated sensor feedback to automate behaviors and improve overall system reliability
- Tested and debugged hardware-software integration to ensure consistent performance under project constraints

SKILLS

Languages: C, C++, Python, MATLAB, Java, VBA, JavaScript, TypeScript, HTML, CSS,
Tools: GitHub, Visual Studio Code, React, Arduino IDE, Tinkercad, Fritzing, OpenCV, MediaPipe
Hardware: Arduino, Microcontrollers, sensors, breadboarding, Embedded Systems, Circuit Design, PCB Design